Analytical Problem Solving

PROBLEM

Question 1 – Problem

Title : Water Jug Problem Using State Space Search

Problem Statement

You are given two water jugs with capacities of 4 gallons and 3 gallons, respectively. Neither jug has any measuring marks. A water pump is available to fill the jugs completely. The allowed operations are:

- Fill either jug completely.
- Empty either jug onto the ground.
- Pour water from one jug into the other until either the source jug becomes empty or the destination jug becomes full.

The objective is to obtain exactly 2 gallons of water in the 4-gallon jug using the minimum sequence of valid operations.

Question 2 – Problem

Title : Mars Rover Intelligent Agent Environment Formulation (PEAS)

Problem Statement

A Mars Rover is an autonomous intelligent agent designed to explore the surface of Mars. It must collect scientific information, analyze rocks and soil, avoid obstacles, and communicate the collected data back to Earth. Using the PEAS (Performance Measure, Environment, Actuators, Sensors) framework, identify the rover's percepts, characterize its operating environment, list the actions it can perform, describe how its performance is evaluated, and determine the most suitable agent architecture.

Question 3 – Problem

Title : 8-Queens Problem Using State Space Search

Problem Statement

The 8-Queens problem requires placing eight queens on an 8×8 chessboard so that no two queens attack each other. This means that no two queens can be placed in the same row, the same column, or the same diagonal. The objective is to formulate the problem components, explore the search space, and determine a valid arrangement using an appropriate AI search strategy.

Question 4 – Problem

Title : OLA Cab Booking Using Goal-Based Problem-Solving Agent

Problem Statement

A customer wants to travel from one location to another using the OLA Cab Booking mobile application. The customer enters the pickup and destination locations and selects a preferred cab type such as Mini, Micro, Sedan, Shared, or Prime. The system searches for available drivers, calculates the fare, and books the cab. Identify the type of problem-solving agent suitable for this application and develop the pseudocode for the booking process.

Question 5 – Problem

Title : Uniform Cost Search (UCS) for Logistics Delivery Network

Problem Statement

A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost representing fuel and time. The delivery network is represented as a weighted graph. Using the Uniform Cost Search (UCS) algorithm, determine the least-cost path from the starting warehouse S to the destination warehouse G, ensuring that the path with the minimum cumulative cost is selected.